SWEEP rule on the KD-Toric code

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1 SWEEP Rule Decoder on the Toric, version II.

add: something that connects between the size of the droplet to the fact it shrinks.

Consider the presentation of a d-dimensional toric by the Cayley graph of the $(\mathbb{Z}_n^d, +)$ group generated by the elements $S = \{1_i : i \in [d]\}$ (where 1_i is the vector consisting of one at the ith coordinate and zero elsewhere). We say that an edge $\{x, y\}$ is positive relative to a plaquette $\{v, g_1v, g_2g_1v, g_2v\}$, defined by a rooted vertex $v \in \mathbb{Z}_n^d$ and the generators $g_1, g_2 \in S$, if $x = v$ and y is either g_1v or g_2v. In our code, bits are set on the plaquette, and the edges correspond to checks that restrict the parity of the bits on their support to be even.

We generalize Toom's rule in the following manner: Each plaquette asks if both of its positive edges point to a syndrome, namely unsatisfied. If so, then the bit flips itself; for convenience, we will think of the vertex at the end of the positive edges as the vertex which makes the decision and flips the plaquette.

Let $\sigma^{(t)}$ be a domain wall at time t, and let $\Gamma^t = \{v : \exists\{v, u\} \in \sigma^t\}$. We denote by ϕ^t the subsets of bits which are flipped as a result of applying the Toom's rule at time t, and by ϕ_v the bits which are flipped by vertex v. We say that u is added to Γ^{t+1} by v at time t, if $u \in \Gamma^{t+1}$ and $u \in V(\phi_v)$. When it clear form the context, we will omit the time, and say that u is added by v.

Claim 1.1. Let $v \in \Gamma^t$ for any u that is added by v, and for any $x \in [d+1]$, we have $L_x(u) \leq L_x(v) + 1$.

Proof. By definition, any u that is added by v is of the form $v + \sum \alpha_j 1_j$, where $\alpha_j \in \{0, 1\}$ and $\sum \alpha_j \leq 2$. Thus, for any $x \in [d]$:

$$L_x(u) = \langle 1_x, v + \sum \alpha_j 1_j \rangle = \langle 1_x, v \rangle + \alpha_x \leq L_x(v) + 1$$

For $x = d + 1$ we have:

$$L_x(u) = \langle -\mathbf{1}, v + \sum \alpha_j 1_j \rangle = \langle -\mathbf{1}, v \rangle - \sum_j \alpha_j \leq L_x(v)$$

□

Claim 1.2. Let $v \in \Gamma^t$ such that $L_{d+1}(v)$ is a (local) maxima in Γ^t then for any u which is added by v we have $L_{d+1}(u) \leq L_{d+1}(v) - 1$.

Proof. We will prove that $v \notin \Gamma^{t+1}$, and then by repeating on claim 1.1 where $\sum \alpha_j \geq 1$, we get the desired result. Assume, toward contradiction, that $v \in \Gamma^{t+1}$, and denote by w the vertex that adds v. If $w \neq v$, then there is at least one coordinate i for which $w_i < v_i$, thus $L_{d+1}(w) > L_{d+1}(v)$, in contradiction to $L_{d+1}(v)$ being a maximum in Γ^t. In the other case, where v adds itself, it follows that σ_v can't be decomposed into pairs of unsatisfied checks; namely, $|\sigma_v|$ is odd. We show by induction that impossible,namely that $|\sigma_v|$ is always even. The induction is over the size of the plaquettes subset that induce the boundary σ.

1. Base. Consider a boundary that induced by a single plaquette, in that case it obvious that $|\sigma_v|$ is either zero (where v is not belong to the plaquette) or 2.

2. Assume for any boundary that induced by at most k plaquettes, and consider a boundary σ which is induced by $k+1$ plaquettes, name this subset F and observes that F can be decomposed to a sum of two disjoint subsets A and B, each with less than $k+1$ plaquettes, Now:

$$\begin{aligned} \sigma_v &= (\partial A)_v \cup (\partial B)_v \\ \Rightarrow |\sigma_v| &= |\,(\partial A)_v\,| + |\,(\partial B)_v\,| - 2|\,(\partial A)_v \cap (\partial B)_v\,| = \text{even} \end{aligned}$$

   Where we used the induction assumption, which tells that $|\,(\partial A)_v\,|$ and $|\,(\partial B)_v\,|$ are even numbers.

□

Claim 1.3. Let $x \in [d]$ and let $v \in \Gamma^t$ such that $L_x(v)$ is a (local) maxima in Γ^t then for any u which is added by v we have $L_x(u) \leq L_x(v)$.

Proof. Since $L_x(v)$ is maximal, v doesn't see edges that goes from it at the positive x-direction, otherwise denote it by e and denote by w the second vertex at the end of e, w has an x-coordinate which is greater than v's x-coordinate by 1 and in addition $w \in \Gamma^t$ (since it lays on the support of e). Therefore $L_x(w) > L_x(v)$ in contradiction to $L_x(v)$ be maximal in Γ^t. Hence, any plaquette that is being flipped by v lies on edges associated with 1_j such that $j \neq x$. Thus, any vertex u that is added by v has the form $v + \sum_{j \in [d] \setminus \{x\}} \alpha_j 1_j$, where $\alpha_j \in \{0, 1\}$ and $\sum \alpha_j \leq 2$. So:

$$L_x(u) = \langle 1_x, v + \sum_{j \in [d]/x} \alpha_j 1_j \rangle = \langle 1_x, v \rangle = L_x(v)$$

□

Claim 1.4. Let Γ^t be the vertices in the support of the domain wall σ^t. Denote $v_i = \arg\max_{v \in \Gamma^t} L_i(v)$. Then for any $i \in [d]$, one can find $u \in \Gamma^{t-1}$ such that u adds v_i and $L_i(u) \geq L_i(v_i)$. In addition, one can find $u_{d+1} \in \Gamma^{t-1}$ such that u_{d+1} adds v_{d+1} and $L_{d+1}(u_{d+1}) = L_{d+1}(v_{d+1}) + 1$.

Proof. Just take u_i to be any vertex that adds v_i. By claims claim 1.1, claim 1.2, and claim 1.3, we get the claim. □

[COMMENT] $V(\phi_v)$ was not defined. $v \leq w$ also not defined.

2 Space-Time Graph

Definition 2.1. Consider a graph $G = (V, A, F)$, where V is a set of vertices, and A, F are sets of undirected edges. For $v \in V$, denote by $\text{pre}(v) = \{u \in V : \{u, v\} \in A\}$, we extend the notion for subsets of vertices $X \subset V$ by $\text{pre}(X) = \bigcup_{v \in X} \text{pre}(v)$. Let $T : V \to \mathbb{N}$, We say that (G, T) is a time graph if and only if:

1. $\{x, y\} \in A \Rightarrow T(y) = T(x) \pm 1$
2. For $x, y \in V$ $\{x, y\} \in F$ if and only if there exists $z \in V$ such both $\{x, z\}, \{y, z\} \in A$.

We will call to T time-function, to A arrows, and to F forks. Observes that from the second requirement it follows that forks can connect only vertices that have the same time value (set by T).

Definition 2.2 (History, slice.)**.** For a time graph (G, T) and a vertex $u \in V$, let G_u be the subgraph of (V, A) induced by $\{v \in V : T(v) \leq T(u)\}$. We will denote by H_u the 'History of u', which is the set of vertices that are connected to u in G_u. A slice is an equivalence class of the relation $u \equiv v$ if and only if $H_u = H_v$. Notice that the time function has to be constant on a slice, to see this consider vertices x, y with $T(x) > T(y)$, thus $x \notin G_y$ and therefore $x \in H_x / H_y$ So they are belong to different classes. Thus, we can extend the time function and the history to slices and write for a slice K, $T(K)$ and H_K.

Claim 2.3. Consider slices K_1, K_2 such that $T(K_1) = T(K_2)$, then either $K_1 = K_2$ or H_{K_1}, H_{K_2} are disjoint.

Proof. Assume a non-empty intersection, so there exists $z \in H_{K_1} \cap H_{K_2}$ such that for any $x \in K_1$ and $y \in K_2$, there exist arrow-paths $z \to x$ and $z \to y$. Thus, for any $w \in H_{K_1}$, one can concatenate a path $w \to x \to z \to y$ and conclude that $y \in H_{K_2}$. □

Definition 2.4 (The graph of slice)**.** Let K be a slice. The graph of K, denoted by $G_K = (V_K, E_K)$, is the graph whose vertices are slices $R \subset H_K$ with $T(R) = T(K) - 1$. Two vertices are connected by an edge if and only if there is a fork whose ends belong to the slices associated with them. Namely, $S, R \in V_K$ are connected in G_K if and only if there is $\{s, r\} = f \in F$ such that $s \in S$ and $r \in R$.

Claim 2.5. G_K is connected.

Proof. Let $R, S \in V_K$ and consider $r, s \in V$ such that $r \in R, s \in S$. Recall that $R, S \subset H_K$ and therefore $r, s \in H_K$, Thus, by definition of the 'History' there exists a path, passing trough arrows only, from r to s in H_K. Consider such a path that is minimal in the number of points x belong to it, with $T(x) = T(K)$. By induction on the number k of such points we prove that R and S are connected in G_K.

1. Base. $k = 0$, In this case, the path from r to s lies within $G_r (= G_s)$. Hence $H_r = H_s$, So $R = S$.

2. Induction step. There exists y on the path from r to s such that $T(y) = T(K)$. Let x be the point which precedes y on the path, and z the point that follows y. Since the values of T at the two ends of an arrow differ by 1, and since y gets the maximal T value in H_K it follows that:

   (a) $\{x, z\} \subset \text{pre}(y) \Rightarrow \{x, z\} \in F$.

   (b) The slices of x and z, say K_x and K_z has time value $T(K_x) = T(K_z) = T(K) - 1$, Namely they are both members of V_K.

   Since the path connecting r, x in H_K has $k - 1$ vertices with $T(\cdot) = T(K)$, then by the induction hypothesis, R and K_x are connected in G_K, and by similar argument so are K_z and S. On the other hand, $\{x, z\} \Rightarrow \{K_x, K_z\} \in E_K$. Thus, R and S are connected in G_K.

□

Definition 2.6. Let $f : G \hookrightarrow \mathbb{R}^d$ be an embedding that maps each element of $V \cup A \cup F$ to a point in $\mathbb{R}^d$. We say that $\left(G, \mathbb{R}^d\right)$ is a space-time graph if and only if:

1.

2. The point set by f for an edge is the mean of its endpoints' images; that is, for $e = \{x, y\} \in A \cup F$, we have $f(e) = \frac{1}{2}\left(f(x) + f(y)\right)$.

Definition 2.7. Define $\text{pre}_i(v) = \arg\max_{u \in \text{pre}(v)} L_i(f(u))$

Claim 2.8. Let $\mathbf{S} = \{S_1, S_2, .., S_k\}$ where $S_i \subset \mathbb{R}^d$, introduce an edge between S_i and S_j iff $S_i \cap S_j \neq \emptyset$. Assume that $\mathbf{S}$ is connected. Then $\mathbf{Size}(\cup_{i=1}^k S_i) \leq \sum_{i=1}^k \mathbf{Size}(S_i)$.

Proof. It suffices to prove that if $R_1 \cap R_2 \neq \emptyset$, then $\mathbf{Size}(R_1 \cup R_2) \leq \mathbf{Size}(R_1) + \mathbf{Size}(R_2)$. Let $a_{i,j} = \max_{v \in R_j} L_i(v)$. Then for any $w \in R_1 \cap R_2$ we have:

$$\begin{aligned}
\mathbf{Size}(R_1 \cup R_2) &= \sum_i^{d+1} \max a_{i,1}, a_{i,2} = \sum_i^{d+1} a_{i,1} + a_{i,2} - \min a_{i,1}, a_{i,2} \\
&\leq \sum_i^{d+1} a_{i,1} + a_{i,2} - L_i(w) = \mathbf{Size}(R_1) + \mathbf{Size}(R_2) - \mathbf{Size}(w) \\
&= \mathbf{Size}(R_1) + \mathbf{Size}(R_2)
\end{aligned}$$

□

Definition 2.9 (Spanned Set)**.** Consider an embedding $f : \star \hookrightarrow \mathbb{R}^d$, a subset of f's domain $S \subset \star$, and $d+1$ points $x_1, \ldots, x_{d+1}$. We say that $\langle S, x_1, \ldots, x_{d+1}\rangle$ is a spanned set of S if for any $i \in [d+1]$ and $s \in S$, it holds that $L(f(s)) \leq L_i(x_i)$ and in addition there are $v_1, ..v_{d+1} \in V$, not necessary different, that achieve the equalities $L_i(x_i) = L_i(f(x_i))$. We name $x_1, \ldots, x_{d+1}$ the poles of S. Furthermore, we extend the notation to a spanned slice when S is a slice of some time-graph. Notice that saying $\langle S, x_1, \ldots, x_{d+1}\rangle$ is a spanned set means that $\mathbf{Size}(S) = \mathbf{Size}(\{x_1, \ldots, x_{d+1}\})$.

Observe that in the case where the subset S is a slice. Each of these poles can be associated with a vertex of G that gives the equality. If there are multiple vertices whose images by f give equality, associate the point with one of them arbitrarily. We will use the notion of applying functions originally defined over vertices to the poles, referring to the application over the vertices associated with them. For example $\text{pre}(x_i) = \text{pre}(v)$ for the vertex $v \in S$ which satisfies $L_i(f(v)) = L_i(x_i)$ and therefore is associated with the pole x_i. Another example is referring to x_i as a vertex, so when saying 'For $x_i \in V_k$' or 'For $\{x_i, u\} \in F$', we refer to the associated vertex or the edge connecting it to u.

Claim 2.10. Consider a time graph (G, T), an embedding $f \colon G \to \mathbb{R}^d$, such that.. , Let K be a slice in G, and let $\hat{K} = \langle K, x_1, ..x_{d+1} \rangle$ be a spanned slice of K such that $K \neq H_K$. Then for some integer k there exist spanned slices $\hat{K}_0 = \langle K_0, \cdot \rangle, .., \hat{K}_k = \langle K_k, \cdot \rangle$ and Forks $F_1, .., F_k$ such that:

1. For $i \in [d+1]$ $\text{pre}_i(x_i)$ belongs to the poles $\hat{K}_0, .., \hat{K}_k$.
2. Introduce an edge between $\hat{K}_i$ and $\hat{K}_j$ if and only if one of the Forks $F_1, .., F_k$ consists of a pole of $\hat{K}_i$ and a pole of $\hat{K}_j$. Then the graph formed from $\hat{K}_0, .., \hat{K}_k$ is connected.
3. $\sum_{i=0}^{k} \textbf{Size}(\hat{K}_i) + \sum_{i=1}^{k} \textbf{Size}(F_i) \geq \textbf{Size}(\hat{K}) + 1$

Proof. First Notice that K doesn't contain leaves in (V, A), since a leaf is connected only to itself via arrows, and therefore its 'slice' equals to its 'History'. Thus $\text{pre}_i(x_i)$ is not empty for $i = 1, .., d+1$. Next consider the graph $G_K = (V_K, E_K)$ from definition 2.4. Since $\{x_i, \text{pre}_i(x_i)\}$ is an arrow $\text{pre}_i(x_i)$ belongs to a slice in V_K. for $i = 1, .., d+1$ denote by R_i the slice in V_K that contains $\text{pre}_i(x_i)$.

By claim 2.5 G_K is connected and therefore there exist a minimal collection $\{K_0, .., K_k\}$ of slices form V_K which contains $R_1, R_2, ..R_{d+1}$, and which is connected in G_K. We pick a set of forks $\mathbf{F} = \{F_1, .., F_k\}$ which realizes a spanning tree for this collection of slices. Later we will identify edges from this spanning tree with the forks from $\mathbf{F}$.

For $i = 1, .., d+1$ we set $\text{pre}_i(v_i)$ to be the ith pole of R_i, This assures (1). The set of remaining poles for $\hat{K}_0, .., \hat{K}_k$ will be equal to $\cup_{i=k}^{k} F_i$. This will assure (ii), We will also select poles for $F_1, .., F_k$ to form 'spanned forks' $\hat{F}_1, \cdots, \hat{F}_k$ in such way that:

$$\sum_{i=0}^{k} \textbf{Size}(\hat{K}_i) + \sum_{i=1}^{k} \textbf{Size}(\hat{F}_i) \geq \sum_{i=1}^{d+1} L_i\left(\text{pre}_i(v_i)\right) \geq \textbf{Size}(\hat{K}) + 1$$

This will assure (3).

Let us denote by $J_0, .., J_{2k}$ the enumerated union $K_0, ..K_k, F_1, .., F_k$. Observes that for any choice of $(d+1)(2k+1)$ elements $z_0^1, z_0^2, z_0^3, .., z_0^{d+1} .., z_{2k}^{d+1}$ in $\mathbb{R}^d$, such that $z_j^i \in J_j$ it holds that:

$$\sum_{i=0}^{k} \textbf{Size}(\hat{K}_i) + \sum_{i=1}^{k} \textbf{Size}(\hat{F}_i) \geq \sum_{j,i} L_i(z_j^i) \tag{1}$$

Later, we are going to take the points $\{z_j^i\}$ from the intersection of J_j's. For convenience, let us assume that for the index subset $X \subset [2k] \cup \{0\}$, the intersection $\bigcap_{j\in X} J_j$ returns a single point. If there is more than one, then decide on the point in a way that all the chosen points simultaneously satisfy that for any index subsets Y, X, such that $X \cap Y$ and they both introduce a non-trivial intersection $\bigcap_{j\in Z} J_j$: $Z \in \{X, Y\}$, then:

$$\bigcap_{j\in X} J_j = \bigcap_{j\in Y} J_j$$

Now consider a non-trivial maximal intersection $\bigcap_{j\in X} J_j$, the intersection is maximal in the sense that for any $j' \notin X$ it holds that $.\bigcap_{j\in X\cup\{j'\}} J_j = \emptyset$. We claim that for any i and for any such X, there is $t \in X$ such that J_t is the successor of J_j on the path to R_i for any $j \in X/\{t\}$. Assume not. Let $x, y \in X$, and denote the paths from J_x, J_y to the ith pole by $\langle J_x, J_{x_2}, \dots, R_i\rangle$, $\langle J_y, J_{y_2}, \dots, R_i\rangle$, such that $J_y \neq J_{x_2}$. If $y = x_2$, $y_2 = x$, or $x_2 = y_2 \in X$, then pick other x, y (if there is no such, we get an immediate contradiction). In that case, $\langle J_x, J_y, J_{y_2}, \dots, R_i\rangle$ is a path from J_x to R_i which is different from $\langle J_x, J_{x_2}, \dots, R_i\rangle$. Therefore, there are at least two paths from J_x to the ith pole, namely, there is a cycle in contradiction to the minimality of $K_0, \dots, K_k$.

We will choose the z_j^i by the following process. Pick a z_j^i that has not been set yet. If $J_j = R_i$, then set $z_j^i \leftarrow \mathrm{pre}_i(x_i)$ - we call this a type-1 assignment. Otherwise, consider the path from J_j to R_i (by connectivity, it has to be such a one, and by minimality, there is exactly one). Let J_t be the successor on the path, then pick $z_j^i \in J_j \cap J_t$ - similarly, we call this assignment a type-2 assignment. Since any type-2 assignment is associated with a non-trivial intersection, and any non-trivial intersection induces a type-2 assignment for all i's (since for all i's, one of the J_j on its support is a successor of the rest on their path to the ith pole), we have that:

$$\begin{aligned}
\sum_{j,i} L_i(z_j^i) &= \sum_{j,i,z_j^i \in \text{type-1}} L_i(z_j^i) + \sum_{j,i,z_j^i \in \text{type-2}} L_i(z_j^i) \\
&= \sum_{i}^{d+1} L_i\left(\mathrm{pre}_i(x_i)\right) + \sum_{X:\, \bigcap_{j\in X} J_j \neq \emptyset} (|X|-1) \sum_{i}^{d+1} L_i\left(\bigcap_{j\in X} J_j\right) \\
&= \sum_{i}^{d+1} L_i\left(\mathrm{pre}_i(x_i)\right)
\end{aligned}$$

Plugging it into eq. (1) gives the desired. □

3 Explantation of a Spanned Slice

[COMMENT] Here we follow the original proof, but change the triminalogy, m will be the number of leaves in the explanation, then in the end we conclude that the number of faults is at least m/Δ. So $|R| \le f(\frac{m}{\Delta})$.

Definition 3.1. Consider a time graph (G, T), an embedding $f: G \to \mathbb{R}^d$, such that.. Let $\beta = \frac{d+1}{\xi}$ and $\alpha = \beta\textbf{Size}(\text{fork})$. Given sets W of points, R of arrows and F of forks, we define U as the set of vertices of the graph induced by the edges of $R \cup F$. The sets W, R and F form an explanation of a spanned slice $\hat{K} = \langle K, v_1, .., v_{d+1}\rangle$ if and only if:

1. $W \cup \{v_1, .., v_{d+1}\} \subset U \subset H_K$ and the graph $(U, R \cup F)$ is connected.
2. All elements of W are leaves.
3. For $m = |W|$ and $s = \textbf{Size}(K)$ we have $|R| \leq \alpha(m-1) - s$ and $|F| < m - 1$.

Claim 3.2. Every spanned slice $K = \langle K, v_1, .., v_{d+1}\rangle$ has an explanation.

Proof. Let $h = T(K) - \min_{u \in H_K} T(u)$. The proof is by induction on h.

1. Base: $h = 0$, namely T is constant on H_K, so no arrows connect points of H_K. Thus, H_K, as well as $\mathbf{K}$, consists only of leaves. In that case, $U = W = \{u\}$, $m = 1$, $s = 0$, and $|R|, |F| = 0$ satisfy the requirement.
2. Induction Step: $h \neq 0$ implies $K \neq H_K$. Thus, by claim 2.10, we know that for some k there exist spanned slices $\hat{K}_0, .., \hat{K}_k$ and forks $F_1, .., F_k$ that satisfy claim 2.10.

   Consider $i \in 0, .., k$, denote by $s_i = \textbf{Size}(\hat{K}_i)$. By the inductive hypothesis and since $T(K_i) = T(K) - 1$, we know that $\hat{K}_i$ has an explanation formed by a set of leaves $\mathbf{W}_i$, a set of arrows $\mathbf{R}_i$ and a set of forks $\mathbf{F}_i$. Let $m_i = |\mathbf{W}_i|$, So the sets R_i and F_i have at most $\alpha m_i - s_i$ and $m_i - 1$ leaves. We define:

   (a) $\mathbf{W} = \cup_{i=0}^k \mathbf{W}_i$, Notice that $\mathbf{W}_i \subset K_i$ and that $T(K_i) = T(K_j)$ for all i, j, thus by claim 2.3 $\mathbf{W}$ is a union of disjoint set, Hence W contains $m = \sum_{i=0}^k m_i$ leaves.

   (b) $\mathbf{F} = \{F_1, .., F_k\} \cup_{i=0}^k \mathbf{F}_i$ with at most $k + \sum_{i=0}^k m_i - 1 = m - 1$ elements.

   (c) $\mathbf{R} = \{\{v_i, \text{pre}_i(v_i)\}\} \cup_{i=0}^k \mathbf{R}_i$ with at most

   $$\begin{aligned} & d + 1 + \sum_{i=0}^{k} \alpha(m_i - 1) - \beta s_i \\ \leq & d + 1 + \alpha m - \alpha(k+1) - \beta\left(s + \xi - k \cdot \textbf{Size}(\text{fork})\right) \\ = & \alpha\left(m-1\right) - \beta s + \left(\alpha + d + 1 - (k+1)\alpha - \beta\xi + \beta k \cdot \textbf{Size}(\text{fork})\right) \end{aligned}$$

   So its enough to show that for $k \geq 0$ the term in the closure is negative, We do that by showing that the slop is negative, and then the value of the term at zero is negative:

   $$-\alpha + \beta\textbf{Size}(\text{fork}) \leq 0 \Rightarrow \textbf{Size}(\text{fork}) \leq \frac{\alpha}{\beta}$$

   $$d + 1 - \beta\xi \leq 0 \Rightarrow \frac{d+1}{\beta} \leq \xi$$

And those inequalities hold for $\beta = \frac{d+1}{\xi}$ and $\alpha = \beta\textbf{Size}(\text{fork})$. Finally, by claim 2.10, the graphs induced by $\mathbf{R}_i \cup \mathbf{F}_i$ are connected, thus the graph induced by $\mathbf{R} \cup \mathbf{F}$ is also connected. Therefore, $\mathbf{W}, \mathbf{R}, \mathbf{F}$ form an explanation of K.

□

4 Counting Subtrees.

Claim 4.1. Let $G = (V, E)$ be a $\Delta + 1$-regular graph, and let k be an integer less than $|E|$. Fix a vertex $r \in V$. The number of subgraphs rooted at r with less than k edges is at most α^k.

Proof. Denote by T_Δ and T_2 the infinity Δ and binary tress. There is an injection between the subtrees of G, rooted at v, with at most k edges, to the subtrees of T_Δ rooted at v, with at most k edges. Moreover, there is an injection from the subtrees of T_Δ rooted at v, with at most k edges to the subtrees of T_2, rooted at v, with at most $k \log \Delta$ edges.

[COMMENT] In the second map, we think on replacing each of the vertex with $\Delta - 1$-leaves binary tree, each leaf is associated with a outgoing edge. .

Thus, it's enough to bound the number of of subtrees in T_2 rooted at v, with at most $k \log \Delta$ edges, that number is less than the number of binary tress with $k \log \Delta + 1$ leaves. For exact leaves number we get the $k \log \Delta$ Catalan number, which its limit is:

$$\sim \frac{4^{k \log \Delta}}{(k \log \Delta)^{\frac{3}{2}} \sqrt{\pi}} \leq \frac{\Delta^{2k}}{(k \log \Delta)^{\frac{3}{2}} \sqrt{\pi}}$$

Therefore we can bound by:

$$\sum_{j=1}^{k} \frac{\Delta^{2j}}{(j \log \Delta)^{\frac{3}{2}} \sqrt{\pi}} \leq \frac{\Delta^{2k}}{\sqrt{\pi} \log^{\frac{3}{2}} \Delta} \sum_{j=1}^{k} \frac{1}{j^{\frac{3}{2}}} \leq \zeta(\frac{3}{2}) \cdot \frac{\Delta^{2k}}{\sqrt{\pi} \log^{\frac{3}{2}} \Delta}$$

□

Now, from here we get an upper bound on the number of subgraphs that connected l vertices at the top level. We bound it from above by counting combination of rooted subtrees, such the vertices are their root and the overall edges number is summed up to k. Thus:

$$\leq \zeta(\frac{3}{2}) \binom{k-1}{l-1} \frac{\Delta^{2k}}{\sqrt{\pi} \log^{\frac{3}{2}} \Delta}$$

5 Tree Separator Lemma

For a tree $T = (V, E)$ and a vertex $v \in V$, we call the subtree obtained by taking the descendants of v the subtree rooted at v.

Claim 5.1. Let $T = (V, E)$ be a tree with $m > 3$ leaves and τ edges, then it has a subtree with $m' \in (\frac{1}{3}m, \frac{2}{3}m)$ leaves and at most $\frac{m'}{m}\tau$ edges.

Proof. Order the vertices according to their heights. Denote by $v \in V$ the lowest vertex for which the subtree rooted at v has at least $\frac{1}{3}m$ leaves. Now, denote by T_1 the subtree obtained through the following iterative process over v's children:

1. Initialize T_1 as the subtree rooted at v.
2. For a child u of v:
   (a) Check if erasing from T_1 the subtree rooted at u doesn't decrease the number of leaves below $\frac{1}{3}m$. If so, then we update T_1 to be the result of the erasure.

Clearly, the algorithm is defined such that T_1 has to have at least $\frac{m}{3}$ leaves. Furthermore, since any child of v is a root of a subtree with fewer than $\frac{m}{3}$ leaves (otherwise we get a contradiction to the minimality of v's height), we have that if T_1 would have more than $\frac{2}{3}m$ leaves, then the subtraction of any of its children would yield a tree with at least $\frac{2}{3}m - \frac{1}{3}m = \frac{1}{3}m$ leaves. Namely, v has in T_1 a child u such that the subtraction of the subtree rooted at u from T_1 results in a subtree with more than $\frac{1}{3}m$ leaves. But if indeed there were such a child, then the algorithm would erase it, so there is not.

Now let $T_2 = T/T_1 \cup v$, namly the tree induced by the subtraction of T_1's edges from T. Observes that since the summation of leaves in T_1 and T_2 is the leaves in T, and T_1 has no more than $\frac{2}{3}m$ leaves, and no less than $\frac{1}{3}$ leaves, it holds that the number of leaves in T_2 is also in the range $(\frac{1}{3}m, \frac{2}{3}m)$. We claim that at least one of the subtrees has fewer than $\frac{m'}{m}\tau$ edges.

To see this, denote by m_1, m_2 and by τ_1, τ_2 the leaves and the edges in T_1, T_2 respectively. Now:

$$\frac{\tau}{m} = \frac{\tau_1 + \tau_2}{m_1 + m_2} = \frac{m_1}{m_1 + m_2}\frac{\tau_1}{m_1} + \frac{m_2}{m_1 + m_2}\frac{\tau_2}{m_2}$$

Namely, $\frac{\tau}{m}$ is a weighted average of $\frac{\tau_1}{m_1}$ and $\frac{\tau_2}{m_2}$, and therefore it is greater than their minimum. Peak the subtree with the minimal quotient. □

By repeating recursively on the above construction, we get the following corollary:

Corollary 5.2. Let $T = (V, E)$ be a tree with m leaves and τ edges. Then, for any $\alpha > 0$, it has a subtree with $m' \in \left(\frac{1}{3}\alpha m, \frac{2}{3}\alpha m\right)$ leaves and at most $\frac{m'}{m}\tau$ edges.

Claim 5.3. Let $\gamma, \beta \in \mathbb{R}_{>0}$ and n be a positive integer. For any tree T with $\tau \geq \beta n$ edges and m leaves such that $\tau \leq \gamma m$, there is a subtree T' of T with a number of edges τ' less than $\frac{\beta}{2}n$ and a number of leaves greater than $\frac{\beta}{4\gamma}n$.

Proof. Since the construction of claim 5.1 preserves the ratio of edges to leaves, we have that for a T' obtained using the construction, the number of edges satisfies $\tau' \leq \gamma m'$. So, it's enough to look for the maximal α such that:

$$\gamma m' \leq \gamma \frac{2}{3}\alpha m \leq \frac{\beta}{2}n \Rightarrow \alpha = \frac{3\beta}{4\gamma}\frac{n}{m}$$

Which gives $m' \geq \frac{1}{3}\alpha m = \frac{\beta}{4\gamma}n$.

$\square$

Move this definition to the beginning of the paper.

Definition 5.4. We model the qubits on the finite toric with a side length n by the projection:

$$(x_1, x_2, .., x_d) \mapsto (x_1 \bmod n, x_2 \bmod n, .., x_d \bmod n)$$

Rewrite when we talk about the first time we have deviation clusters of size $\in (\alpha d, d)$. The reason is that having those assumptions means all the droplets in the past shrank; thus, we can use the ratio of edges to time-leaves in the big subgraph.

Claim 5.5. Assume that until step τ, there was no deviation cluster of size greater than d. Let $\hat{K}$ be a spanned slice induced by a deviation of a cluster state at step τ, and let ω be an explanation for $\hat{K}$ such that the number of vertices in ω is greater than βn. Then there is a subgraph (subtree) ω' of ω such that ω' has fewer than $\frac{\beta}{2}n$ edges and at least $\frac{\beta}{4\gamma}n$ leaves.

Proof. Let ω be an explanation of $\hat{K}$ with more than βn vertices. Since there is no deviation cluster of size greater than d until step τ, we have that every vertex of ω is a spanned slice induced by a boundary of a deviation cluster of size at most d. Thus, ω satisfies the shrink property in Claim 1.4, and by Claim 3.2, the ratio of edges to leaves of ω is at most γ. Observe that a spanning tree of ω has a ratio of edges to time-leaves that is at most ω's ratio, and the number of edges equals ω's vertices minus 1. Therefore, by applying claim 5.3 on ω's spanning tree, ω has a subtree with less than $\frac{\beta}{2}n$ edges and at least $\frac{\beta}{4\gamma}n$ time-leaves.

$\square$

Proof. Now, since any two connected vertices are at a space-distance of at most 1, it follows that all the leaves in the subtree are at a space-distance of at most βn from each other. Thus, if $\beta < 1$, no two of them have the same space-coordinate modulo n. And therefore the probability to have such subtree is at most $p^{\frac{\beta}{4\gamma}n}$.

Because any ω with more than βn vertices implies the existence of a subtree with size less than $\frac{\beta}{2}n$, it's enough to bound the probability of having such a subtree. Considering that the trees are invariant to shifting by $n \cdot \mathbf{1}_i$, we have that it's sufficient to show that there are no such trees which are rooted in the finite cube $\mathbb{Z}_n^d \times \mathbb{Z}_t$. Finally, we use claim 4.1 to bound the number of such trees given a fixed root. Overall, we get:

$$\mathbf{Pr}\left[\text{No } \omega\text{s with more than } \beta n \text{ vertices}\right] \leq n^d t \left(\xi^2 p^{\frac{\beta}{4\gamma}}\right)^n$$

$\square$

6 The Turns Idea.

The main problem we have is that, for now, the history is defined only for configurations that, at time t, see a non-trivial syndrome. What we can do is imagine that at each turn we flip the bits according to a fixed order. We think of the basic time unit as a turn, and each real-time iteration contains tn^d turns. Now, if a bulk of bits A is flipped at time t, then there must be a turn $\tau \leq tn^d$ such that at that turn there was a configuration that has a history.

Now we would like to describe what the history of that configuration looks like, and we do that by connecting the turn τ to the closest $t'n^d$. The idea is that we cann't ensure the shrinking lemma. Yet we can tell for sure that if there is expansion then its limited.

I have also the felling that the shrinking works, and the idea goes as follow, assume that $v \in \Gamma^{(t-1)}$ gets the maximum of L_i either we stop before flipping a bit that is 'supported' on v or not, in the first case, we get that any u that is added by v' might has $L_i(u) = L_i(v') < L_i(v)$, in the second case we have already handled.

The turn machinery makes sense only in the finite case. Yet, we can back to the Tree Separator Section, and show that with probability

$$n^d \cdot tn^d q^n$$

We don't have a big trees which implies independence. Then we can show that with probability:

$$n^d t(n^d)^2 q'^{\delta n}$$

There was no turn in which it had a configuration that was connected at the time, saw a syndrome, and had a side-length that behaved like $\Theta(n)$.

[COMMENT] The model has to redefined again such it will be clear that outcomes are possibles running of the system. Currently this can not be inferred.

Informally, a quantum circuit is a placement of a collection of quantum gates, taken from a finite set, in different space-time locations. Each space-time location, associated with a gate, encodes the time of the application and the qubits in the support of the gate. Formally

Definition 6.1 (Quantum Circuit)**.** A space-time location l is a tuple that encodes in its first coordinate time, and in its second coordinate a subset of integers referring to a support of (q)bits. For a collection of quantum gates $\mathcal{G} = \{g \colon L(\mathcal{H}_2^{\otimes \Delta}) \to L(\mathcal{H}_2^{\otimes \Delta})\}$, and integers $w, d \in \mathbb{N}$ a quantum circuit C, with depth d and width w, is a collection of tuples $\{u_i = (g_i, l_i)\}$, where u_i are the gates of C and each u_i is composed of a gate $g_i \in \mathcal{G}$ and a space-time location $l_i \in \mathbb{Z}_d \times \mathbb{Z}_w^{\Delta}$. The gates of C have to satisfy the constraint that for every two gates with the same time coordinate, there cannot be a non-empty intersection in their space coordinate, namely for $u_i = (g_i, l_i), u_j = (g_j, l_j) \in C$, the equality $l_{i,1} = l_{j,1}$ implies $l_{i,2} \cap l_{j,2} = \emptyset$. We define the t-cut of C to be $C|_t := \{u_i = (g_i, l_i) \colon u_i \in C \text{ and } l_{i,1} = t\}$.

To define the running of a quantum circuit C on a given (mixed) quantum state, we will need to have the notion of the acyclic directed graph associated with C.

Definition 6.2 (Associated Computation DAG)**.** Let $C = \{u_i\}$ be a quantum circuit with depth d and width w. We define the associated computation DAG of C, denoted by π_C, to be the acyclic directed graph $\pi_C = (\mathcal{U}, E)$ as follows:

1. The vertex set $\mathcal{U}$ is the gate set of C, namely $\mathcal{U} = \{u_i\}$.
2. For $u_i = (g_i, l_i), u_j = (g_j, l_j)$, There is an edge $\{u_i, u_j\} \in E$ if and only if:
$$j \in \arg\min_j \{l_{j,1} : l_{j,1} > l_{i,1} \text{ and } l_{j,2} \cap l_{i,2} \neq \emptyset\}$$
Note that there might be multiple indices that get the minimum, so the out degree might be greater than one.

In addition, suppose that $J = \{j_1, j_2, \ldots, j_k\} \subset [|C|]$ is an order of the indices of the gates in C according to the topological order induced by π_C. Then, we say that J is an order of indices that agrees with π_C and define the series of circuit $C_J = \{C_{J,i}\}$ as follows:

1. The first element: $C_{J,1} = \{u_{j_1}\}$
2. Then, $C_{J,i+1} = C_{J,i} \cup \{u_{j_i}\}$. We name $C_{J,i}$ the partial computation circuit until step i.

Definition 6.3. Let ρ be a density matrix over w qubits, and let $C = \{u_i\}$ be a quantum circuit with depth d and width w. For any density matrix σ, quantum gate $g \in \mathcal{G}$, and location l, the application of the gate-location tuple (g, l) on σ, denoted by $(g, l) \circ \sigma$, is the application of g on σ where wires are ordered such that the first qubit in the input to g is $l_{2,1}$, the second is $l_{2,2}$, and so on. The result of the application of C on ρ, denoted by $C \circ \rho$, is given by iterative applications of the gates $\{u_i\}$ on ρ, according to a topological order induced by π_C.

We say that $\rho_1, \rho_2, \ldots, \rho_k$ is a computation prefix if $\rho_1 = \rho$ and $\rho_{i+1} = u'_i \circ \rho_i$, where $\{u'_i\}$ is a prefix of a topological ordering of $\{u_i\}$. If $\{u'_i\}$ contains all the gates in $\{u_i\}$, then we call $\rho_1, \rho_2, \ldots, \rho_k$ a running.

Definition 6.4 (Associated Computation DAG)**.** Let $C = \{u_i\}$ be a quantum circuit with depth d and width w. For a set of fault locations $\mathcal{E} \subset [d] \times [w]$, we define the quantum circuit $C[\mathcal{E}] = \{u'_i\}$ to be the result of the following map:

$$u_i = (g_i, (l_{i,1}, l_{i,2})) \mapsto \begin{cases} (g_i, (2l_{i,1}, l_{i,2})) & u_i \in C \\ (g_i, (2l_{i,1}-1, l_{i,2})) & u_i \in \mathcal{E} \end{cases}$$

We will denote the associated computation DAG of C affected by $\mathcal{E}$ as $\pi[C, \mathcal{E}]$.

Definition 6.5 (Faults Propagation to a Step τ)**.** For a quantum circuit $C = \{u_i\}$, with depth d, an order $J = \{j_1, j_2, \ldots, j_k\} \subset [|C|]$ that agrees with π_C, and gate u_i in C, a propagation of a gate $u_i \in C$ to step τ is the quantum operator $X\colon L(\mathcal{H}_2^w) \to L(\mathcal{H}_2^w)$ obtained by the follow process:

1. First, check if i precedes τ in J. If not, then return $X = I$. Otherwise, denote $j_l = i$, and look for an operator X_1 such that $u_{j_l} u_{j_{l+1}} = u_{j_{l+1}} X_1$.
2. Then, for $k \leq \tau - l$, look for an operator $X_k \colon L(\mathcal{H}_2^w) \to L(\mathcal{H}_2^w)$, such that $X_{k-1} u_{j_{l+1}} = u_{j_{l+1}} X_k$.
3. At the final stage of the process, namely where $k = \tau - l$, we define $X := X_k$ to be the propagation of u_j to step τ.

For a quantum circuit C, a set of faults $\mathcal{E} = \{e_i\}$, an order $J \subset [|C[\mathcal{E}]|]$ that agrees with $\pi_{C[\mathcal{E}]}$, and a step $\tau \in J$, denote by $X_{e_i}^\tau$ the propagation of $e_i \in \mathcal{E}$ in $C[\mathcal{E}]$ to step τ. Let $j'_1, j'_2, \ldots, j'_l$ be the indices of the faults in $\mathcal{E}$ such that their corresponding positions in J precede τ, ordered according to the reverse topological order induced by $\pi_{\mathcal{E}}$. The propagated error to step τ is defined by

$$X^\tau := \prod_i X_{e_{j'_i}}^\tau$$

Definition 6.6 (Deviation)**.** For a quantum circuit C, a set of faults $\mathcal{E}$, and step τ, let X^τ be the propagated error at step τ. We define the deviation of $C[\mathcal{E}]$ from C at step τ by the set of indices on which X^τ acts non-trivially.

Definition 6.7. Using the same notations as in Definition 6.6, for a set of unitary operators $\mathcal{S} = \{s \colon \mathcal{H}_2^w \to \mathcal{H}_2^w\}$, we define the propagated error up to stabilizers in $\mathcal{S}$ as $X^\tau[S] := \min_{s \in \mathcal{S}}\{w(sX)\}$. Then, we define the deviation of $C[\mathcal{E}]$ from C at step τ up to stabilizers in $\mathcal{S}$ to be the subset of coordinates on which $X^\tau[S]$ acts non-trivially.

Definition 6.8 (Generalized Tanner Graph)**.** For registers a and b, their generalized Tanner graph is their 'union' $\mathcal{T}_a \cup \mathcal{T}_b$. This allows us to define the Tanner graph of a quantum circuit C, at step τ, by stacking together Tanner graphs. This also allows us to define the Tanner graph if a register. rewrite. add definition for a register R of quantum circuit C.

Definition 6.9 (Clustered Deviation)**.** For a quantum circuit C, a register R of C, and a set of faults $\mathcal{E}$, let $\mathcal{T}_R$ be the Tanner graph of R, and let $\mathcal{D}_\tau$ be the deviation of $C[\mathcal{E}]$ from C on R at step τ. Denote by $\mathcal{T}_R|_{\mathcal{D}_\tau}$ the subgraph of $\mathcal{T}_R$ induced by the restriction of the left vertices (bits) to the indices in $\mathcal{D}_\tau$. Let us abuse and extend the connection notion to Tanner graphs by defining that two left vertices are connected if and only if they share a right neighbor. We define the deviation clusters at step τ to be the connected components of $\mathcal{T}_R|_{\mathcal{D}_\tau}$. The size of each deviation cluster is the number of right vertices it contains.

Claim 6.10. Let C be a quantum circuit composed of gates with bounded fan degree $\Delta \in \mathbb{N}$. In addition, let R be a quantum register encoded by a quantum error correction code $\mathcal{C}$ with Tanner graph $\mathcal{T}$ that has left and right degrees Δ_1 and Δ_2. Fix a set of faults $\mathcal{E}$, and an order J that agrees with $\pi_{C[\mathcal{E}]}$. For any τ, denote by $Y_1^{(\tau)}, Y_2^{(\tau)}, \ldots, Y_l^{(\tau)}$ the deviation clusters at step $\tau \in J$. Assume that at step $\tau_1 \in J$

there is a deviation cluster $Y_i^{(\tau_1)}$ of size $|Y_i^{(\tau_1)}|$. Then there is a deviation cluster at step $\tau_1 - 1$, let's denote it by Z, such $Z \cap Y_i^{(\tau_1)} \neq \emptyset$ and Z is of size at least:

$$|Z| \geq \frac{1}{\Delta\Delta_1\Delta_2}|Y_i^{(\tau_1)}|$$

Furthermore there is a deviation cluster at step $\tau_1 - 1$, let's denote it by Z', such $Z' \cap Y_i^{(\tau_1)} \neq \emptyset$, and Z' is of size at most:

$$|Z'| \leq \Delta\Delta_1\Delta_2|Y_i^{(\tau_1)}|$$

Add: that when it isn't mentioned, we assume that the error at time t can be decomposed into a product $\Rightarrow$ at each step, a fault is supported on at most one (q)bit.

Proof. Denote by $u = (g, l)$ the gate at step τ. If u acts trivially on the (q)bits in $Y_i^{(\tau_1)}$ then they also belong to deviation on register R at step $\tau_1 - 1$, Since they are connected in $\mathcal{T}$ they form a deviation cluster on register R at step $\tau_1 - 1$ and we got the desired. Thus it is left to handle the case in which u acts non-trivaily on (q)bits in $Y_i^{(\tau_1)}$. Since u acts on at most Δ qubits, and each qubit has at most $\Delta_1(\Delta_2 - 1)$ neighbors at $\mathcal{T}$ it follows that u acts non trivialy on at most

$$\Delta + \Delta\Delta_1 (\Delta_2 - 1) \leq \Delta\Delta_1\Delta_2$$

deviation clusters at step $\tau_1 - 1$. Let l be integer no greater than $\Delta\Delta_1\Delta_2$ and let $Y_1, Y_2, \ldots, Y_l$ be the deviation cluster, on register R, at step $\tau_1 - 1$, on which u acts non trivialiy, since $Y_i^{(\tau_1)} \subset \bigcup_j Y_j$. We have that:

$$\begin{aligned} |Y_i^{(\tau_1)}| &\leq \sum_j |Y_j| \leq \Delta\Delta_1\Delta_2 \max_j |Y_j| \\ \Rightarrow \max_j |Y_j| &\geq \frac{1}{\Delta\Delta_1\Delta_2}|Y_i^{(\tau_1)}| \end{aligned} \tag{2}$$

Complete the second inequality. Should consider to erase what have already written about it. To get the second inequality, we use the fact that u is a unitary gate, in particular, u is reversible. Let $C_{J,\tau_1} = u_1, .., u_{\tau_1}$ be the partial computation circuit until step τ_1. Observe that $C_{J,\tau_1 - 1}$ is obtained by stacking $u_{\tau_1}^\dagger$ as a final stage to C_{J,τ_1}, denoted by $u^\dagger \circ C_{J,\tau_1}$. Thus, the deviation from C_{J,τ_1} at step $\tau_1 - 1$ equals the deviation from $u^\dagger \circ C_{J,\tau_1}$ at step $\tau_1 + 1$. By eq. (2), □

Corollary 6.11. Let $\alpha = 1/(\Delta\Delta_1\Delta_2)$. Assume that at the initial step j_1 there is no deviation cluster of size more than αd, and that at a step $\tau_1 \in J$ there is a deviation cluster $Y_i^{(\tau_1)}$ of size $|Y_i^{(\tau_1)}| > d$. Then there is a step $\tau_o < \tau_1, \tau_o \in J$ for which there is a deviation cluster Z at step τ_o, such that the size of Z is in the range $(\alpha d, d)$.

Proof. □

Claim 6.12. Let A be the event in which, until time t, there is a time t' such that there is a deviation cluster of size greater than d. For any τ, let B_τ be the event that step τ is the first step to has a deviation cluster of size in the range $(\alpha d, d)$. Then:

$$\mathbf{Pr}\left[A\right] \leq 2wd \max_{\tau\in[2wd]} \mathbf{Pr}\left[B_\tau\right]$$

Proof.

$$\mathbf{Pr}\left[A\right] \leq \mathbf{Pr}\left[\bigcup_{\tau} B_\tau\right] \leq 2wd \max_{\tau\in[2wd]} \mathbf{Pr}\left[B_\tau\right]$$

□

7 Tanner - History Connected Equivalency

Imagine the following process, we apply t noisy iterations of the SWEEP rule, and then a single ideal (without noise) iteration. Denote by ξ a deviated configuration (can be either bits or checks), and by ξ' the result of applying an ideal SWEEP rule on ξ. Notice that if ξ is connected in the Tanner graph then ξ' is connected in the History graph **[COMMENT]** Require proof, For the classical case enumerate all the options can be done by hand. Thus:

$$\mathbf{Pr}\left[\xi \text{ connected in Tanner}\right] \leq \mathbf{Pr}\left[\xi' \text{ connected in Time} \mid \text{no noise at the } t+1 \text{ iteration}\right]$$
$$\leq \sum_{m}^{\infty} |\{\text{ explanation to } \xi' \text{ over } m \text{ leaves restricted to no noise at the final iteration }\}| p^{\frac{m}{\Delta}}$$
$$\leq \sum_{m}^{\infty} |\{\text{ explanation to } \xi' \text{ over } m \text{ leaves }\}| p^{\frac{m}{\Delta}}$$
$$\leq \mathbf{Pr}\left[\xi' \text{ connected in Time } \textbf{- Upper Bound}\ \right] \leq q^{\mathbf{Span}(\xi')} \leq q^{\mathbf{Span}(\xi)-1}$$

7.1 Another Idea.

Reconsider the construction of the explanation. We could add at the top an imaginary layer in which all the vertices at time t are connected by arrows to all the vertices (at time $t+1$) that are adjacent to them in the Tanner graph. For the last layer, we might find that the span of a slice expands instead of shrinking. Namely, claim 2.10 for the final slices would change to:

$$\sum_{i=0}^{k} \mathbf{Size}(\hat{K}_i) + \sum_{i=1}^{k} \mathbf{Size}(F_i) \geq \mathbf{Size}(\hat{K}) - \xi$$

Thus, following the computation at the induction stage in claim 3.2, the number of arrows is bounded by:

$$\begin{aligned}&\alpha\left(m-1\right) - \beta s + d + 1 + \alpha + \beta\xi \\ &\alpha\left(m-1\right) - \beta s + O(1)\end{aligned}$$

8 Intermediate Step Lemma

[COMMENT] The idea of this step is also to bound the case that all the bits are flipped at once, So checks are satisfied (and the argue above is not applied directly).

Definition 8.1 (Connected Cluster With Span s)**.** Let $G = (B, C, E)$ be a Tanner graph of a code, and let $f: G \to \mathbb{R}^{d+1}$ be an embedding of the graph into $\mathbb{R}^{d+1}$. For $s \in \mathbb{R}$, we say that $X \subset B$ is a connected cluster of G with span s regarding f if X is connected in G (where two bits are connected if there is a check that connects both of them in E) and $f(X) \subset \mathbb{R}^{d+1}$ has $\mathbf{Size}(f(X)) \leq s$. When it's clear from the context which G and f are, we will call X in brief a connected cluster with span s.

Definition 8.2. The model, computation, and noise are tossed, and then a correction is made.

Claim 8.3. Let ξ be the event in which, at the **beginning** of time t, the error contains a connected cluster with a span greater than $\frac{d}{4}$, and for $i \in [t]$, let σ_i be the event in which, at the **end** of time i, there is a connected cluster with a span of at least $\sqrt{d}$ that belongs to the error. Then:

$$\mathbf{Pr}[\xi] \leq t\mathbf{Pr}[\sigma_1] + p^{\frac{1}{8}\sqrt{d}}$$

Proof. By the law of total probability it hodls:

$$\mathbf{Pr}[\xi] = \mathbf{Pr}\left[\xi \cap \bigcup_i^t \sigma_i\right] + \mathbf{Pr}\left[\xi \cap \bigcap_i^t \bar{\sigma}_i\right]$$

First, let us bound from above the probability of the right intersection, In that case, at the **end** of time $t-1$, the erorr consists of clusters each at span less than $\sqrt{d}$. Let m be the number the clusters and denote them by $X_1, .., X_m$. Now since the clusters are disjoint, the sum of their $\mathbf{Size}(X_i)$ is lower than the total number of the bits, namely:

$$\sum_1^m \mathbf{Size}(X_i) \leq n \Rightarrow m \leq \frac{n}{\sqrt{d}}$$
$$p\frac{n}{\sqrt{d}} \leq \sqrt{d}$$

, Thus, . So denote by τ the number of bits that require to be flipped in order to merge the flipped connected component to a one with a span greater than $d/4$, So τ has to be at least:

$$\tau\left(\sqrt{d}+1\right) \geq \frac{1}{4}d \Rightarrow \tau \geq \frac{1}{8}\sqrt{d}$$

Therefore:

$$\begin{aligned}\mathbf{Pr}\left[\xi \cap \cap_i^t \bar{\sigma}_i\right] &\leq \sum_{\xi^{(t-1)}} \mathbf{Pr}\left[\xi | \xi^{(t-1)}\right] \mathbf{Pr}\left[\xi^{(t-1)} \cap \cap_i^t \bar{\sigma}_i\right] \\ &\leq \sum_{\xi^{(t-1)}} p^{\frac{1}{8}\sqrt{d}} \mathbf{Pr}\left[\xi^{(t-1)} \cap \cap_i^t \bar{\sigma}_i\right] \leq p^{\frac{1}{8}\sqrt{d}}\end{aligned}$$

$$\begin{aligned}\mathbf{Pr}\left[\xi\right] &= \mathbf{Pr}\left[\xi \cap \cup_i^t \sigma_i\right] + \mathbf{Pr}\left[\xi \cap \cap_i^t \bar{\sigma}_i\right] \\ &\leq t\mathbf{Pr}\left[\sigma_1\right] + p^{\frac{1}{8}\sqrt{d}}\end{aligned}$$

□

Definition 8.4. The atomic events are the sets of space-time locations in which faults occur. A run ω is the sequence of states (assignment of all the bits) induced by an atomic event (and the transition function of the machine). We say that a run ω closes a non-trivial cycle if it contains a state in which there is a connected cluster A such that its boundary is not a co-boundary.

Define the bits-cluster graphs $G = (V, E)$ such that the vertices are connected bit-clusters at different times $\{A_i^t\}$, and there is an edge between A_i^t and A_j^{t-1} only if $\Pi_x A_i^t \cap \Pi_x A_j^{t-1} \neq \emptyset$. Given a state at time $t-1$ such that no connected component in G forms a closed non-trival circle, consider $\tilde{A}_i^t$ constructed as follows: perform the application of either noise or Toom's correction in an arbitrary order and stop right before $\tilde{A}_i^t$ forms a closed non-trivial circle.

Claim 8.5. There is $\phi \in Aut(\mathbb{T}_n^d)$ for which $\mathbf{Size}(\phi(\tilde{A}_i^t)) \geq n-1$.

Claim 8.6. Consider a running in which no non-trivial circle was closed, then there is $\psi\colon \mathbb{T}_n^d \to \mathbb{Z}_{2n}^d$ such:

1. For any slice K, $\psi(K)$ is connected in $\mathbb{Z}_{2n}^d$.
2. For any t, denote by S_t the state at time t. Then, for any $t > 1$, we have that $\psi(S_t)$ is the image of the application of the Tooms rule over $\psi(S_{t-1})$.

Proof. We prove it by induction on time t.

1. Base case, $t = 1$. We will construct ψ in the following iterative manner. First, let ψ be the identity. Then, while there are slices that are not connected on $\mathbb{Z}_{2n}^d$, do the following: Let K be a slice that is not connected on $\mathbb{Z}_{2n}^d$. Then it can be decomposed into a disjoint union $K = K_1 \cup K_2$ such that K_1 and K_2 are connected on $\mathbb{Z}_{2n}^d$. Assume, without loss of generality, that for a coordinate $x \in [d]$, it holds that for any $a \in K_1$ and $b \in K_2$, we have $b_x - a_x > 1$.

   Observe that there has to be at least one such coordinate; otherwise, it is a contradiction to K_1 and K_2 not being connected in $\mathbb{Z}_{2n}^d$. Then define:

   $$\psi(z) \leftarrow \begin{cases} \psi(z) & z \cap K_1 = \emptyset \\ z + n\mathbf{1}_x & \text{else} \end{cases}$$

   Repeat the above for other coordinates if $\psi(K)$ is still not connected in $\mathbb{Z}_{2n}^d$. The proof that $\psi(K)$ has to be connected (that the process ends) is omitted.

2. Induction step. We repeat the construction in the base case. To prove the second property, we show that ψ maps (un)satisfied edges to (un)satisfied edges. **[COMMENT]** That is wrong.

$\square$

Claim 8.7. Denote by ω running of the above type, then there is a running ω' on $\mathbb{Z}^d_\infty$ such that ω and ω' agree.

Claim 8.8. Consider a t-steps running, conditioned on the event that no trivial-cycle was closed in the $t-1$ first steps, the probability that a non-trivial cycle will be closed at the tth step is less than:

$$\mathbf{Pr}\left[\Lambda^t | \bigcap_{t'<t} \bar{\Lambda^{t'}}\right] < n^{2d} \sum_{i=n-1}^{\infty} p^i \mathcal{J}_i \leq n^{2d} q^{n-1}$$

Proof. Consider the event in which there is a droplet that at time t closes a non-trivial cycle. Let $\tilde{A}$ be the subset of that droplet obtained by the process in claim 8.5. By the claim, $\tilde{A}$ has a boundary σ and two vertices on that boundary $M = \{u, v\}$ such that $\mathbf{Size}(M) > n-1$, up to isomorphism.

Now, since no slice in the investigation of M closes a non-trivial cycle then by claim 8.6 .. so by claim 8.7 .. $\square$

9 Configuration up to $C_z^\perp$

10 Drafts:

In addition since, in a tree, the number of leavs is at least greater than the maximal degree, then we have that each J_j intersects with at most $d+1$ elements.

$$L + \Delta_{\max} + 2(n - L - 1) \leq \sum_{v \in V} \deg(v) = 2(n-1)$$

$$\Rightarrow L \geq \Delta_{max}$$

11 Decoder Phase

In the general picture, I would like to use the 4D toric code to encode in any target code. We use the original fault tolerance construction (the concatenation construction) to encode into and decode from the 4D toric code. So, for the decoding phase, we need to have a decoder which, in the ideal setting, runs in less than logarithmic time.

What we know so far:

1. We can assume that two unsatisfied checks are belong to the same history with probability less than $\sim q^{|u-v|}$.

2. There is a $\log^2(n)$ depth algorithm to find the perfect/max matching in a general graph. From first check, the known algorithms to deal with the weight matching are based on LP.

3. Applying Toom's/Sweep rule will require at least linear depth.

4. Having an algorithm that succeeds with constant probability $\frac{1}{2} + \epsilon$ might be enough (Anyway the qubit at the result of the decoding is going to absorb p-noise).